Supplementary File S2 — PLOS Human Participants Research Checklist

Study: Cross-dataset EEG state classification under proxy labels via multi-source domain generalization (MSDG) with few-shot calibration.

Study type: Computational **secondary analysis** of eight (8) previously published, publicly available, de-identified human EEG datasets. No new participants were recruited and no new primary data were collected by the authors.

#	Disclosure item (PLOS human-participants research)	Response / Detail
1	Does the study involve human participants?	**Yes** — it analyzes human EEG recordings (secondary use of existing data).
2	Ethics approval / IRB obtained?	**Yes** — Youngsan University Institutional Review Board granted an **exemption** (protocol No. **YSUIRB-202607-HR-219-02**; receipt 2026-07-08; confirmed 2026-07-22) for secondary analysis of de-identified public EEG data, under the Republic of Korea *Bioethics and Safety Act* and 45 CFR 46.104(d)(4).
3	Informed consent	**Not applicable (waived)** — all source datasets are de-identified public repositories released under original informed-consent agreements permitting research reuse; this secondary analysis involved no contact with, and no re-identification of, any participant.
4	Participant population	Human volunteers whose EEG was recorded in the original studies. Datasets: DREAMER, DEAP, MAHNOB-HCI, SEED, SEED-IV, FACED, OpenNeuro ds004572, OpenNeuro ds006437 (see manuscript §3.1).
5	Recruitment by the authors	**N/A** — pre-existing public datasets were used; no recruitment was conducted.
6	Eligibility / inclusion criteria	Determined by the original dataset protocols; documented in manuscript §3.1 and §3.2.
7	Sample size / power	Driven by each dataset's original sample; no prospective power calculation (secondary analysis). Cross-dataset benchmarking uses all eligible recordings per manuscript §3.1.
8	Data collection methods	Authors used the original preprocessed/public EEG signals; no new collection. Partitioning follows participant-level GroupShuffleSplit to avoid leakage (manuscript §3.2).
9	Anonymization / privacy	All datasets are de-identified. No names, addresses, DOB, patient/hospital numbers, contact info, IP, location, signatures, financial, or identifiable media are presented. Processed features are regenerated by public code and not separately redistributed (manuscript §3.7).
10	Vulnerable populations	Not specifically targeted; source cohorts are general adult samples per original publications.
11	Reporting guideline adherence	STROBE / TRIPOD+AI-style reporting checklist provided as Supplementary File S1 (`reporting_checklist_S1.pdf`); ethics/IRB item addressed (S1 item 17).
12	Data availability	All underlying data are public (manuscript Data Availability Statement); code at Zenodo https://doi.org/10.5281/zenodo.21531272 (concept DOI; v1.0.13 versioned record https://doi.org/10.5281/zenodo.21922961) and GitHub https://github.com/korose523/Hypnotise (tag v1.0.13).
13	Conflicts of interest	None declared (manuscript Author Contributions).
14	Funding	No specific funding received for this work (manuscript Acknowledgments).

Confirmations

- ☑ Ethics exemption on record (YSUIRB-202607-HR-219-02).
- ☑ No identifiable data presented; all sources de-identified.
- ☑ Secondary analysis only; no new participant contact or consent solicitation.
- ☑ Reporting checklist (S1) and data/code availability statements complete.